



Internship Report

Prediction of RNA 3D structure using gradient-descending and
knowledge-based scoring function

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Internship carried out from 02/03/2026 to 04/09/2026

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Abstract

Write an abstract here.

Key words: financial models, machine learning...

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1 Presentation of the company

2 Preliminary notions

2.1 Central Dogma

Every cells of every living being, excluding virus and a few exceptions, has **DesoxyriboNucleic Acid (DNA)**. This molecule contains the genetic information. It holds all information for an organism. However the DNA alone cannot do anything, it needs to be express in proteins. This process is called the central dogma of molecular biology. For that, the cell perform two steps :

1. Transcription : This steps copy a part of DNA and transcribes it into a intermediate molecule called **RiboNucleic Acid (RNA)**, or more precisely messenger RNA (mRNA). The transcription is performed inside the nucleus.
2. Translation : The RNA obtained in the previous step are now used to make a protein. The translation is performed outside of the nucleus.

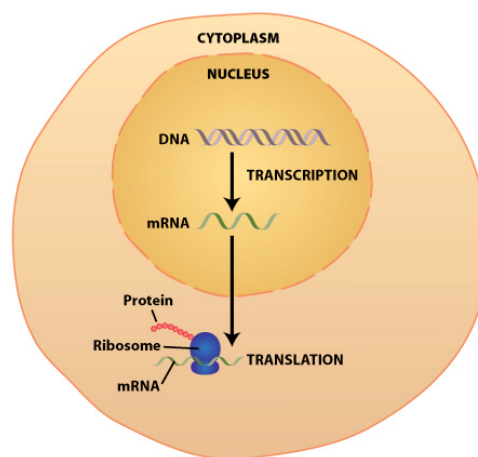


Figure 1: A diagram illustrating the processes of DNA transcription into RNA and RNA translation into proteins. Source : [1]

Recently, researchers have discovered that RNA serves more than just an intermediate role. Indeed, many RNAs, known as non-coding RNAs, are never translated into proteins. Instead, they have a regulation role inside the cell.

2.2 RNA molecule

The RNA is composed composed of nucleotids. The linear arrangement of these nucleotides defines the 1D strcuture or the raw sequence. A nucleotid have three parts :

- Nitrogenous base or nucleobase : There are four type of nucleobase : Adenosine (A), Cytosine (C), Guanine (G), Uracile (U)
- Ribose
- Phosphate group

3 Introduction

4 Data and Dataset

4.1 PDB database

The **Protein Data Bank (PDB)** [2, 3] is the principal worldwide archive for three-dimensional structures of biological macromolecule such as proteins and nucleic acids. This database centralize all structure determined by experimental methods such as X-ray crystallography, nuclear magnetic resonance spectroscopy and cryo-electron microscopy. It can be accessed through its website and an API. Each structure has an identifier composed of four letters or numbers (e.g. 1J8G) and they can be download in two format : PDB file and **macromolecular Crystallographic Information File (mmCIF)** files. A format change happen in 2014, with the advent of cryo-electron microscopy, researchers had the tools to see larger molecules like RNA polymerase. These types of molecules have a lot of atoms but PDB file are restricted to only 99 999 atoms, with mmCIF files, there is no limit on the molecule's scale. As a result, every new molecule added to the database is now in this format. These two format will be describe in the next section.

4.2 PDB file structure

The PDB file is a file to represent each atom from a molecule, especially for protein and RNA. Each line have a keywords to explicite the line content. The file can be split in 3 parts.

The first one contains header and metadata. This part have all information about the molecule and the experimentation.

Keywords	Description
HEADER	Molecule Type, PDB Id, deposition date
TITLE	Title of the experiment
COMPND	Description of the molecule compenent
SOURCE	Biological origin or Synthetic
KEYWDS	Descriptive keywords for the structure
EXPDTA	Experimental Methods
AUTHOR	Author of the deposit
REVDAT	Revision history
JRNL	Publication reference
REMARK	Comments numbered from most important to least important

Table 1: All keywords founded in a first part of a PDB files

The table 1 present all keywords and his signification. REMARK contains useful information such as the resolution of the molecule. The resolution is the difference between the atoms found experimentally and the actual atoms. More the resolution is low, more the experimental molecules can be accurate.

The second part are about the sequence and the crystallography parameters. These keywords are all in the table 2

Keywords	Description
HEADER	Molecule Type, PDB Id, deposition date
TITLE	Title of the experiment
COMPND	Description of the molecule compenent

Table 2: All keywords founded in a second part of a PDB files

SEQRES can be used to find the acid nucleics sequence that can be retrieved and put inside a algorithm to predict the three-dimensional structure.

The last part is the

5 Materials and Methods

5.1 Knowledge based score function

5.2 Objective function

5.3 Optimization algorithm

5.4 Complete Pipeline

5.5 Different Applications

6 Results

6.1 Interpolation results

6.2 Comparaison between original and predicted structures

6.3 Comparatives tables

6.4 Line chart of score function for folding trajectory

6.5 Internet platform for RNA folding

7 Discussion and Perspectives

8 Conclusion

Glossary

D | **M** | **P** | **R**

D

DNA DesoxyriboNucleic Acid. [1](#),

M

mmCIF macromolecular Crystallographic Information File. [2](#),

P

PDB Protein Data Bank. [2](#),

R

RASP Ribonucleic Acids Statistical Potential.

RNA RiboNucleic Acid. [1](#),

References

- [1] Alcohol Pharmacology Education Partnership. Duke University. URL: <https://sites.duke.edu/a pep/module-3-alcohol-cell-suicide-and-the-adolescent-brain/biology-and-chemistry-connections/dna-transcription-translation-synthesis-of-proteins/> (visited on 07/21/2026).
- [2] H. M. Berman et al. “The Protein Data Bank”. In: *Nucleic Acids Research* 28.1 (2000), pp. 235–242. DOI: 10.1093/nar/28.1.235. URL: <https://www.rcsb.org>.
- [3] RCSB Protein Data Bank. “Updated resources for exploring experimentally-determined PDB structures and Computed Structure Models at the RCSB Protein Data Bank”. In: *Nucleic Acids Research* 53 (2025), pp. D564–D574. DOI: 10.1093/nar/gkae1091.